Transition Probabilities in MDP

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User

how we show Transition Probabilities matrix in MDP

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In a Markov Decision Process (MDP), the transition probability matrix is a fundamental component that describes the dynamics of the system. Here's how transition probabilities can be represented:

Components of an MDP

An MDP is defined by:

States (S): A set of all possible states the environment can be in.
Actions (A): A set of all possible actions available to the decision-maker (agent).
Transition probabilities (P): The probabilities of transitioning from one state to another given a specific action.
Rewards (R): A reward function that provides feedback to the agent after taking an action in a given state.
Discount factor (γ): A factor to discount future rewards.

Transition Probability Matrix Representation

The transition probabilities can be represented as a matrix (P), where:

Each matrix element (P(s'|s,a)) indicates the probability of transitioning to state (s') from state (s) after taking action (a).

Structure

If there are (|S|) states and (|A|) actions, then the transition probabilities can be represented in a multi-dimensional array (or tensor) of shape (|S| \times |A| \times |S|).

Example

Let:

(S = {s_1, s_2, s_3}) (3 states)
(A = {a_1, a_2}) (2 actions)

The transition probability (P) can be structured as follows:

For each action (a \in A):

For each state (s \in S):

Define a vector of probabilities for transitioning to any state (s' \in S).

This leads to a representation like:

[P(a_1) = \begin{bmatrix} P(s_1|s_1,a_1) & P(s_2|s_1,a_1) & P(s_3|s_1,a_1) \ P(s_1|s_2,a_1) & P(s_2|s_2,a_1) & P(s_3|s_2,a_1) \ P(s_1|s_3,a_1) & P(s_2|s_3,a_1) & P(s_3|s_3,a_1) \ \end{bmatrix}]

$$ P(a_2) = \begin{bmatrix} P(s_1|s_1,a_2) & P(s_2|s_1,a_2) & P(s_3|s_1,a_2) \\ P(s_1|s_2,a_2) & P(s_2|s_2,a_2) & P(s_3|s_2,a_2) \\ P(s_1|s_3,a_2) & P(s_2|s_3,a_2) & P(s_3|s_3,a_2) \\ \end{bmatrix} $$

Interpretation

Each entry $P(s'|s,a)$ in the matrices sums to 1 for a fixed state s and action a.
This matrix is used in various algorithms, such as value iteration and policy iteration, to evaluate and improve policies in the MDP.

In practice, depending on the complexity of the system, these probabilities can represent deterministic or stochastic transitions among the states.